Computer Organisation

Topic 1

Introduction

(Part 1)

Stallings, Computer Organization & Architecture (9th Ed.), Chap. 1

Outline

- History of computers
- X86, PowerPC and ARM evolution

Organisation and architecture

Structure and function

Computer evolution and performance

Cloud Computing

Performance Issues

Architecture & Organisation

- **Architecture** is those attributes **visible** to the programmer (programmer's view of a computer)
- It is defined by the instruction set
- (language), and operand locations (registers and memory).
  - Deals with the functional behaviour of a computer system
  - Attributes: a system that has a direct impact on the logical execution of a program
  - It defines the system in an abstract manner
  - It deals with <u>What does the system do</u>
  - Instruction set, number of bits used for data representation, I/O mechanisms, addressing techniques
  - e.g. Is there a multiply instruction?

Architecture & Organisation

•Organization is how features are implemented

- Deals with structural relationship
- Attributes: hardware details transparent to the programmer
- Is the way in which a system must structure and It is operational units and the interconnections between them that achieve the architectural specifications
- It is the realization of the abstract model
- It deals with How to implement the system
- Control signals, interfaces between the computer and peripherals, memory technology
- e.g. Is there a hardware multiply unit or is it done by repeated addition?

Architecture & Organisation

Computer architecture is about "what" a computer system does, while computer organization is about "how" it does it. Architecture defines the capabilities and features, while organization determines how those features are implemented in hardware.

- A systematic approach of deriving the solution of any problem
- Manufacturers produce **family** of computer models
  - **Same architecture**, **different organisation**
  - Different models = different price and performance

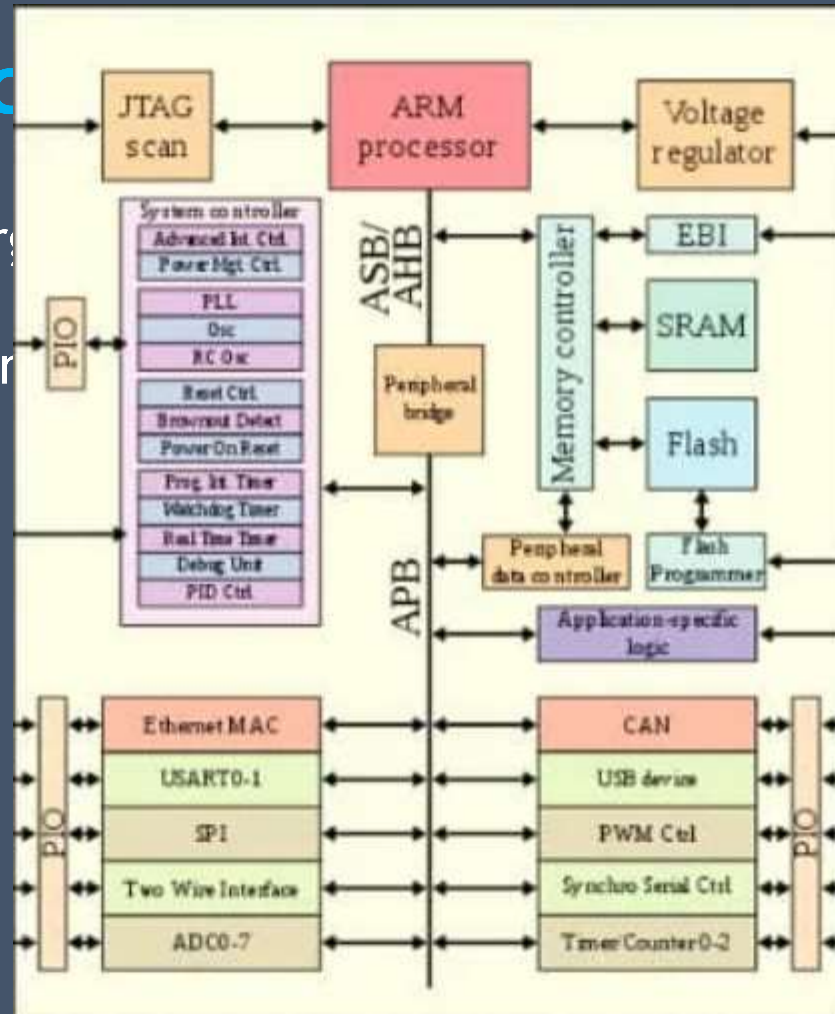
Architecture & Organisation

- All **Intel x86 family** share the same basic architecture
- Similarly **ARM Cortex-M, R and A families** have the same respective ARMv7-M, R and A architectures

sation

x86 PROCESSORS (from Intel)

Bits	Family	Clock Speed (approx.)	Bus Size (bits)	Max RAM	Storage Range	OS
64	Xeon	4.5GHz		3TB	500GB-10TB	Windows: 11, 10, 8, 7 XP, 2000 NT, 98 95, 3.x
	Core i9	5.8GHz	64	128GB		
	Core i3, i5, i7	3.3GHz		64GB		
	Core 2 Duo	2.6GHz				
	Pentium 4	3.8GHz				
	Pentium D	3.4GHz				
32	Core Duo	2.2GHz				Linux Mac OS X SCO Unix Solaris
	Pentium 4	2.8GHz				
	Xeon	3.2GHz				
	Celeron	2.4GHz		4GB	500MB-60GB	
	Pentium III	1.2GHz				
	Pentium II	450MHz				
	Pentium Pro	233MHz		64GB		DOS DR DOS OS/2 Misc DOS Multiuser
	Pentium	200MHz		4GB		
	486DX	100MHz	32		200 - 500MB	
	486SX	40MHz				
	386DX	40MHz			60 - 200MB	
	386SX	33MHz				
	386SL	25MHz				
16	286	12MHz	16	16MB	20-80MB	DOS DR DOS Win 3.x OS/2 1.x
	8086	10MHz		1MB	10-20MB	DOS DR DOS
	8088	5MHz	8			



Outline

- History of computers
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Structure and function

Computer evolution and performance

Cloud Computing

Performance Issues

Structure & Function

- A computer is a **complex** system
  - A **hierarchical** system is a set of interrelated **subsystems**
  - At each level, look into the components and their **interrelationships**

Structure & Function

- **Structure** is the way in which **components relate** to each other
- **Function** is the **operation** of the individual component as part of the structure

Structure & Function

- In terms of description, we have two choices:

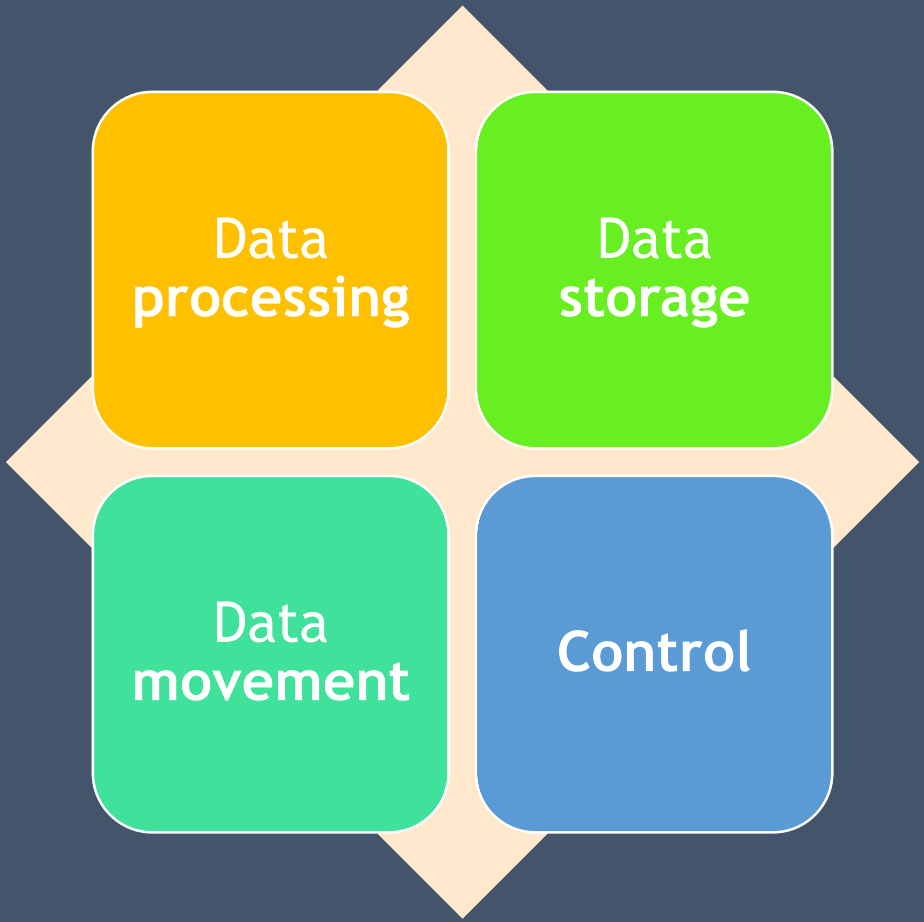
Bottom-up	Top-down
Starting at the bottom and building up to a complete description	Beginning with a top view and decomposing the system into its subparts

Structure & Function

- Clearer to describe computer system using **top-down** approach
  - start with major components (their structure & function)
  - proceed to the lower layer

Function

All computer functions are:

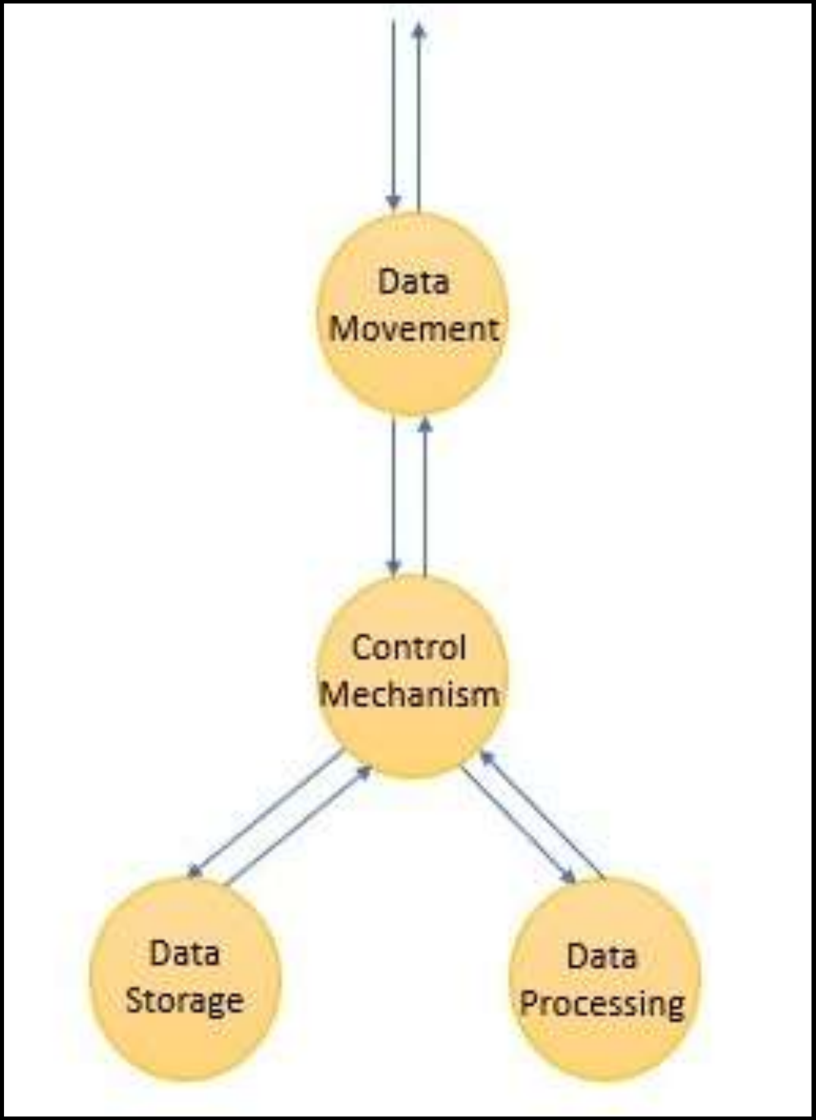


Function

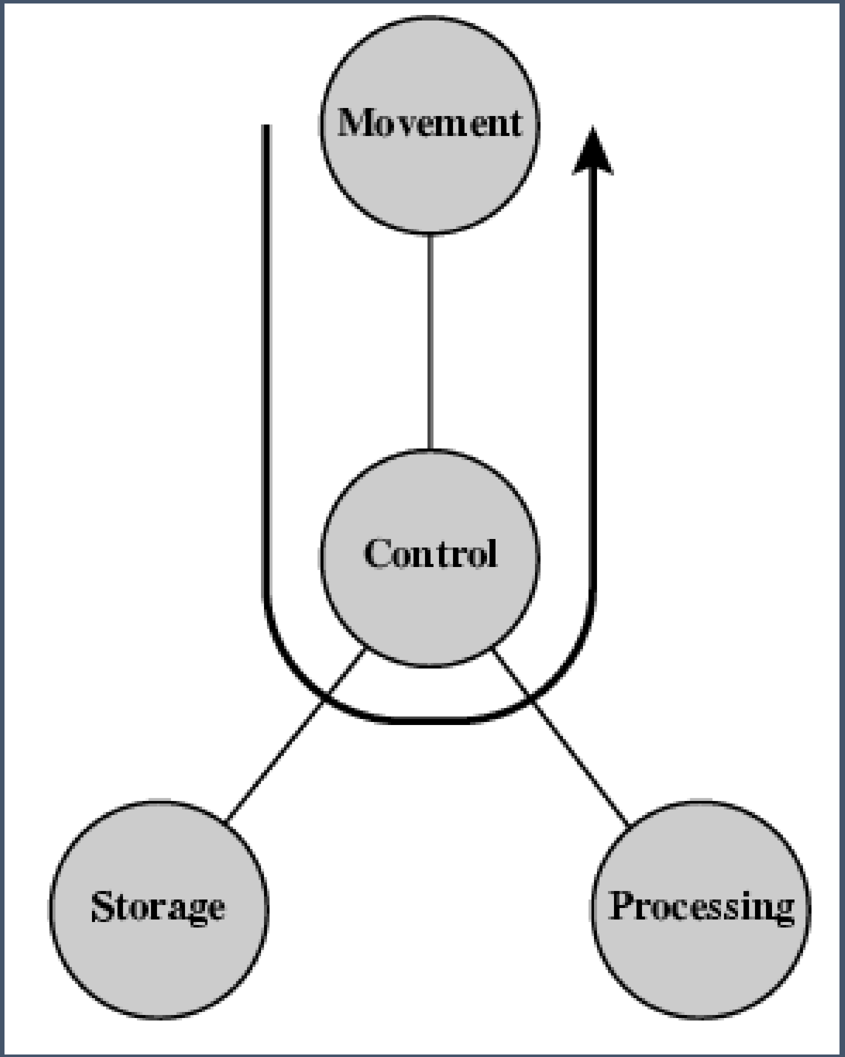
All computer functions are:

Data processing	• Data may take a wide variety of forms and different processing requirements
Data storage	• Even though the computer process the data on the fly, the computer must temporarily store at least those pieces of data • Short term vs long term data storage
Data movement	• Input-output (I/O) Peripheral
Control	• Manages the computer resources

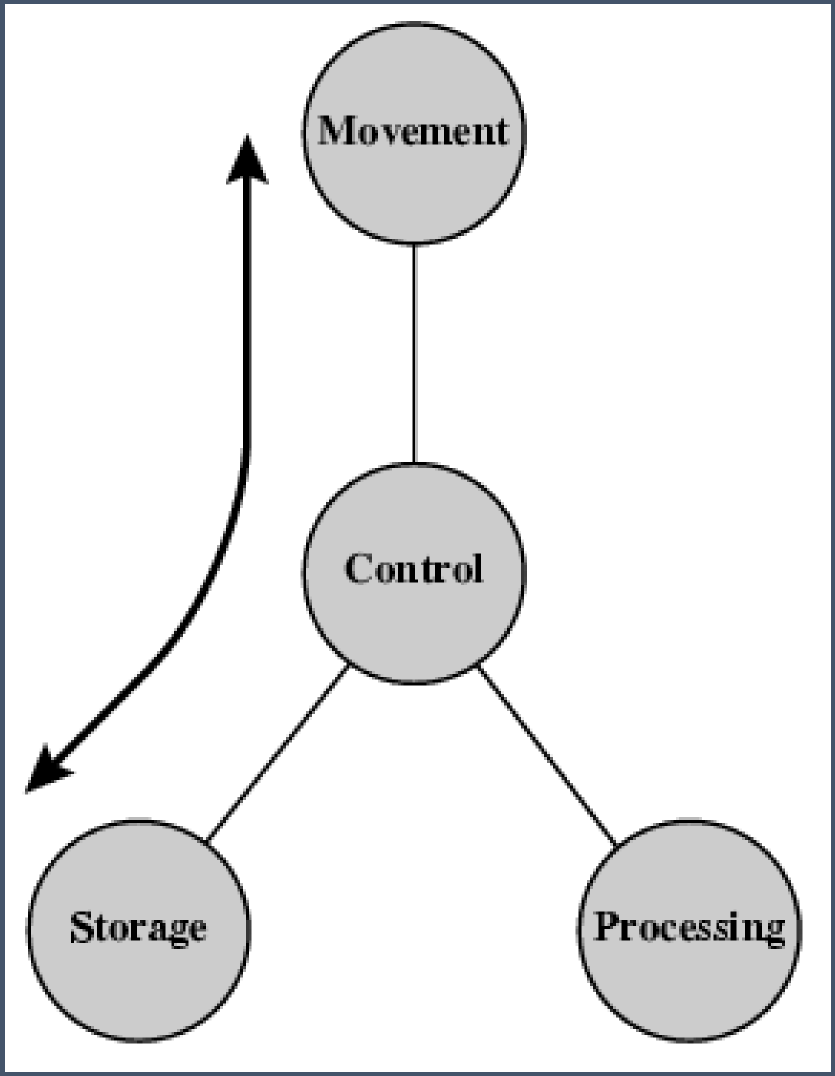
Functional View



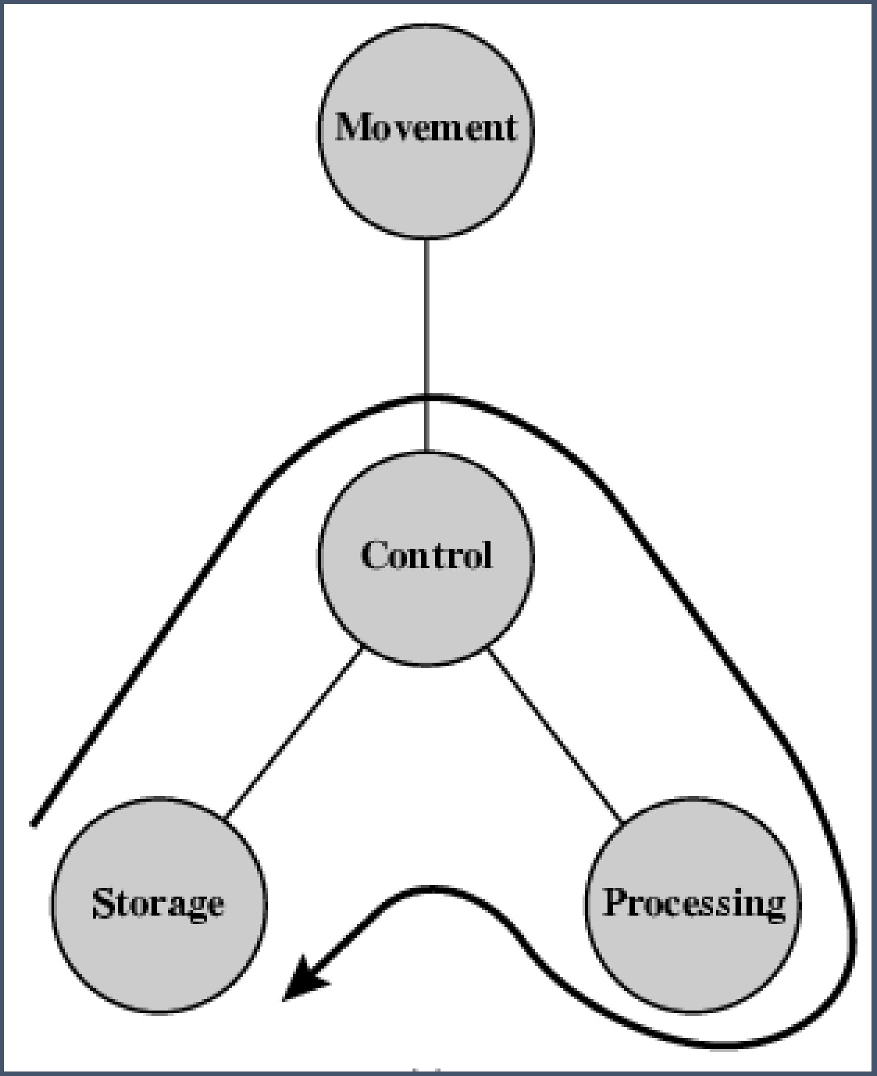
Operations
(a) Data movement



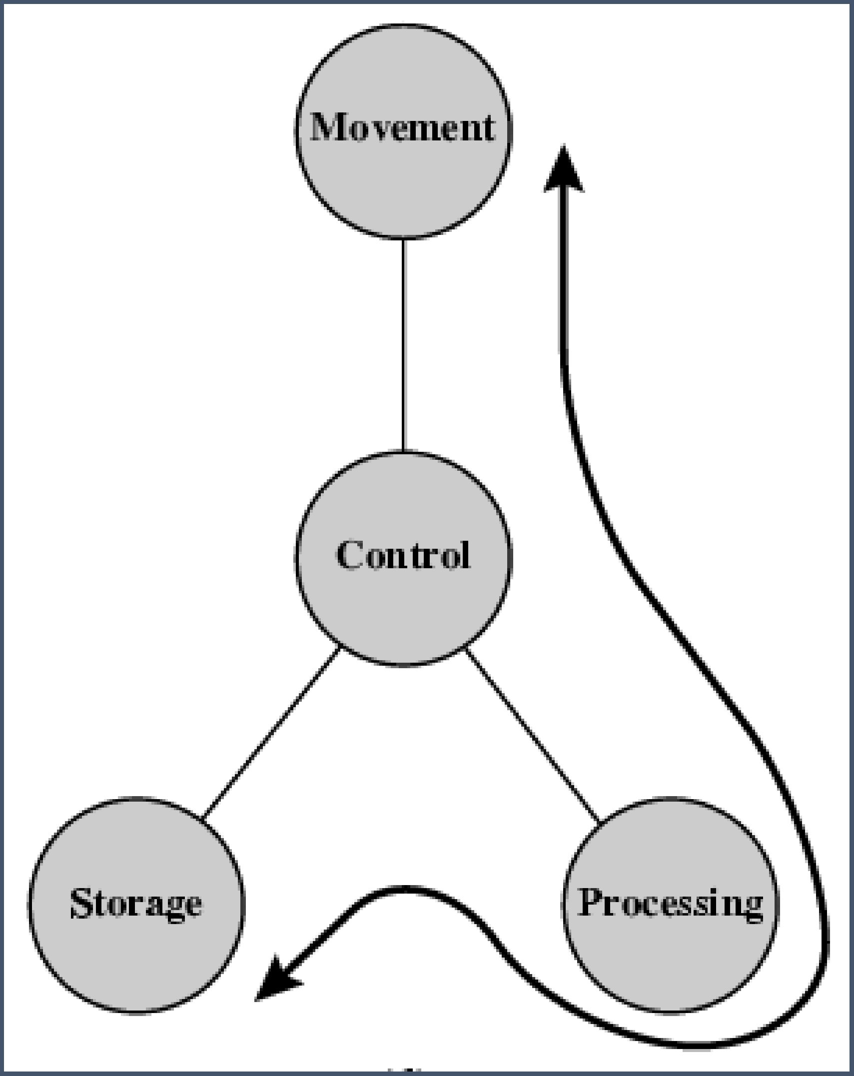
Operations
(b) Storage



Operation
(c) Processing from/to storage

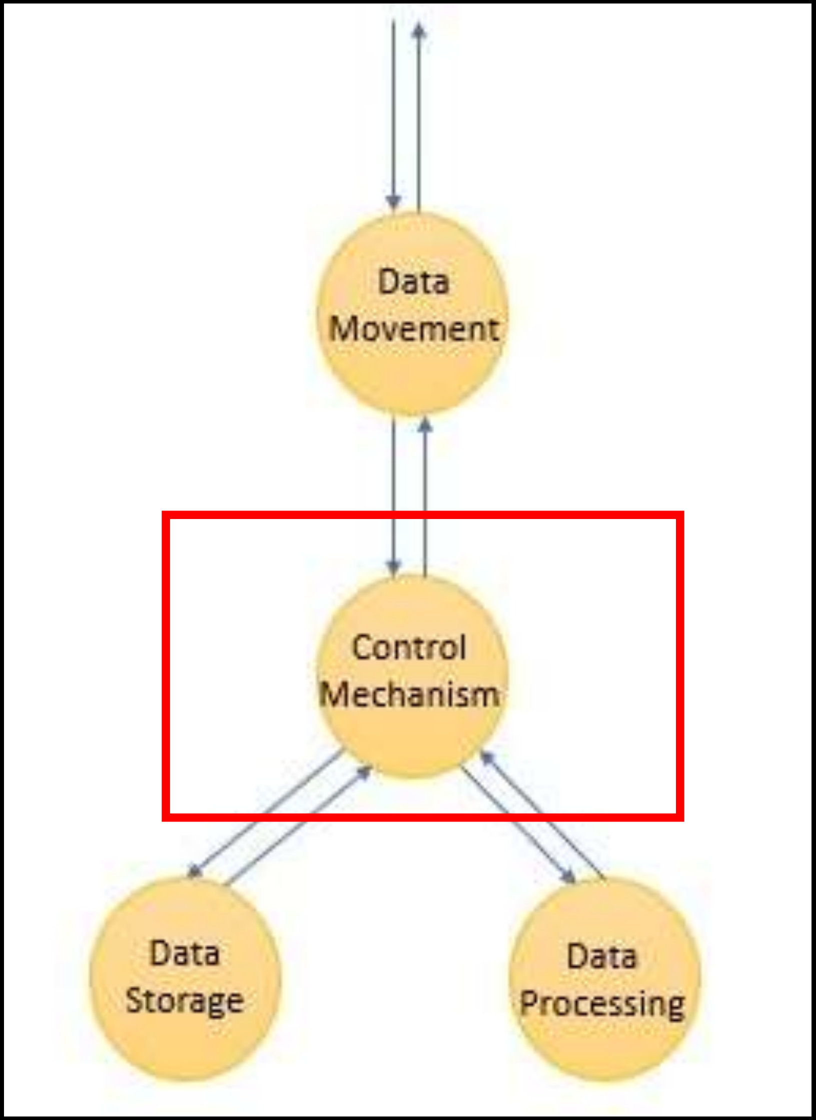


Operation
(d) Processing from storage to I/O



Control

a unit that would control and synchronize the function of these units



Function

- Data processing
- Data storage
- Data movement
- Control

?

- **Which is the most important?**

Function

- Data processing
- Data storage
- Data movement
- Control

- Which is the “busiest”?

Function

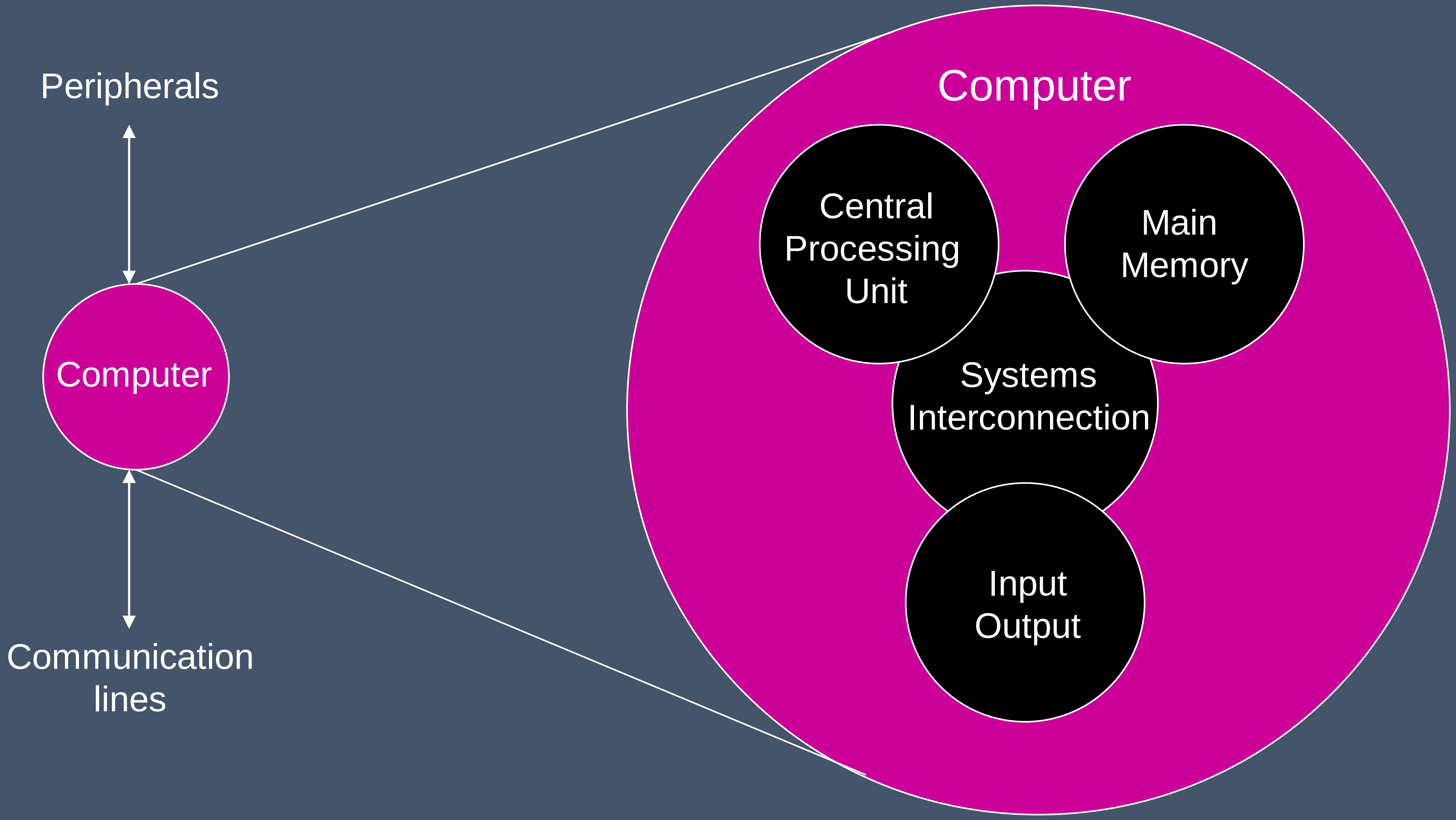
- Data processing
- Data storage
- Data movement
- Control

- So, which is the most important? ☺

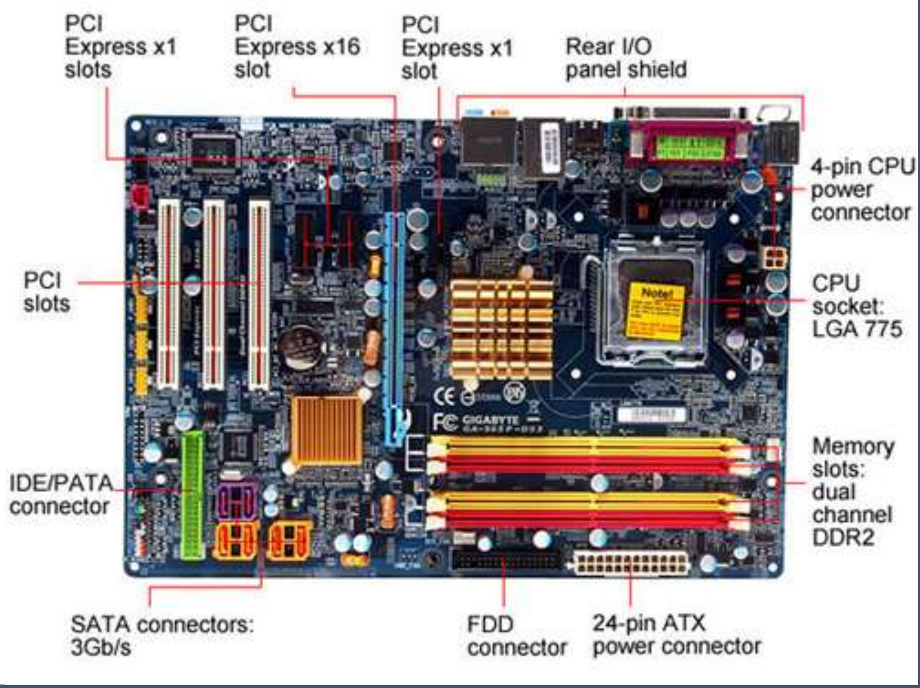
Structure

- Internal structure of a computer:
- 4 main components
  - **Central processing unit** (or processor)
    - controls the operation and performs data processing
  - **Main memory**
    - stores data
  - **I/O**
    - moves data between the computer and its external environment
  - **System interconnection**
    - mechanism for communication among CPU, main memory and I/O

Structure
Top Level



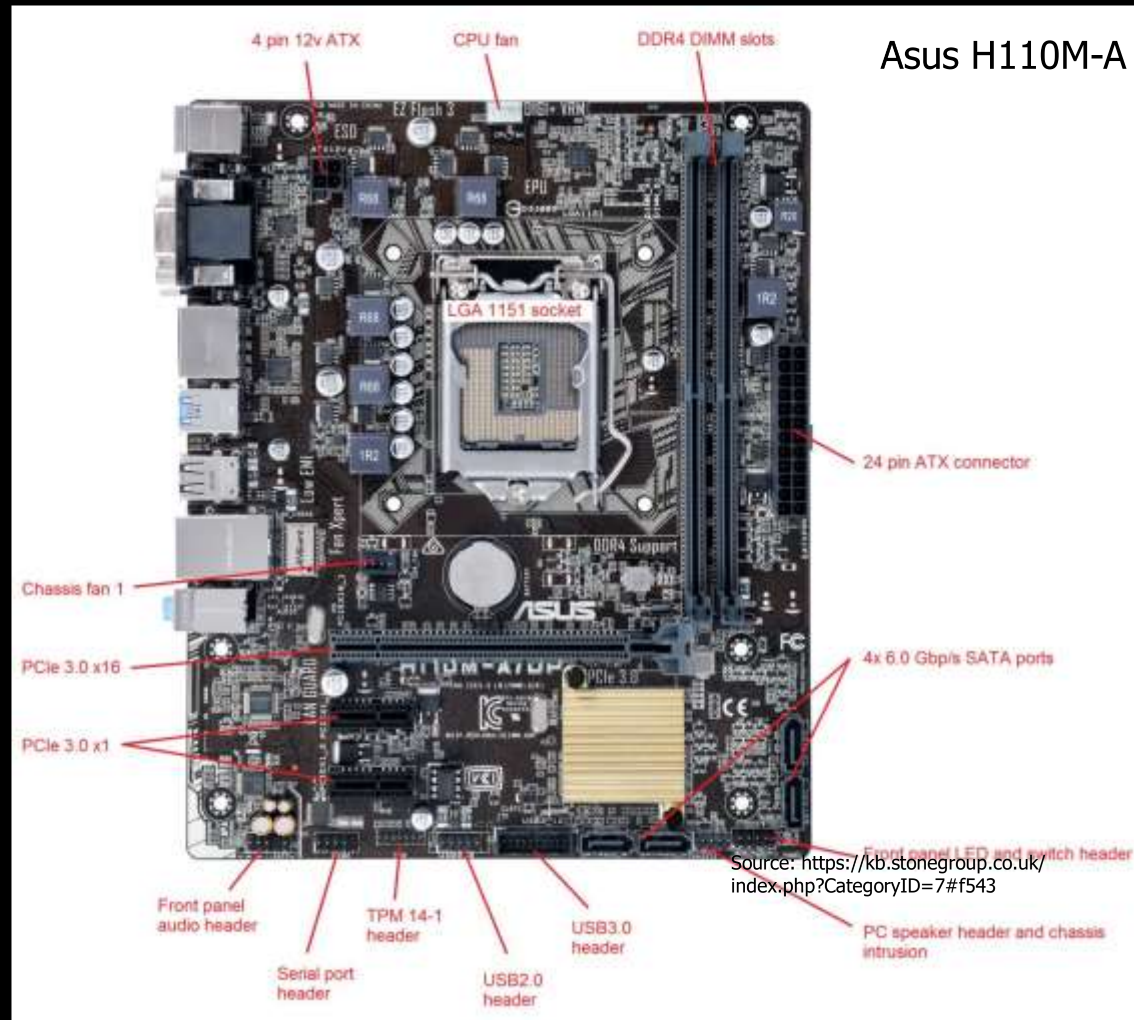
Structure
Top Level (Hardware)



Source: https://www.123rf.com/photo_9645973_black-stylish-desktop-pc-isolated-on-white-background.html

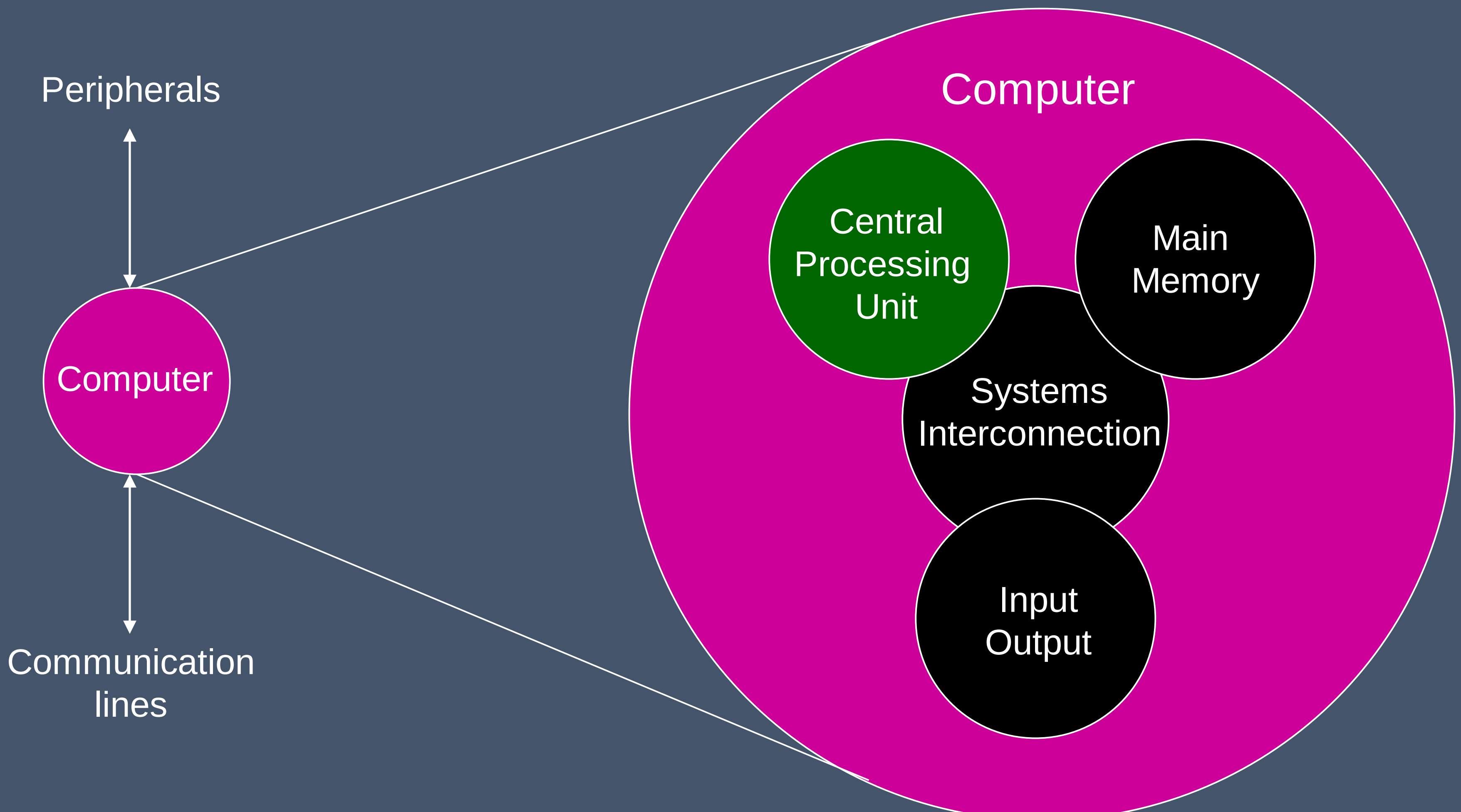
Source: https://www.legitreviews.com/gigabyte-ga-965p-ds3-motherboard-review_418

Structure Top Level (Hardware)

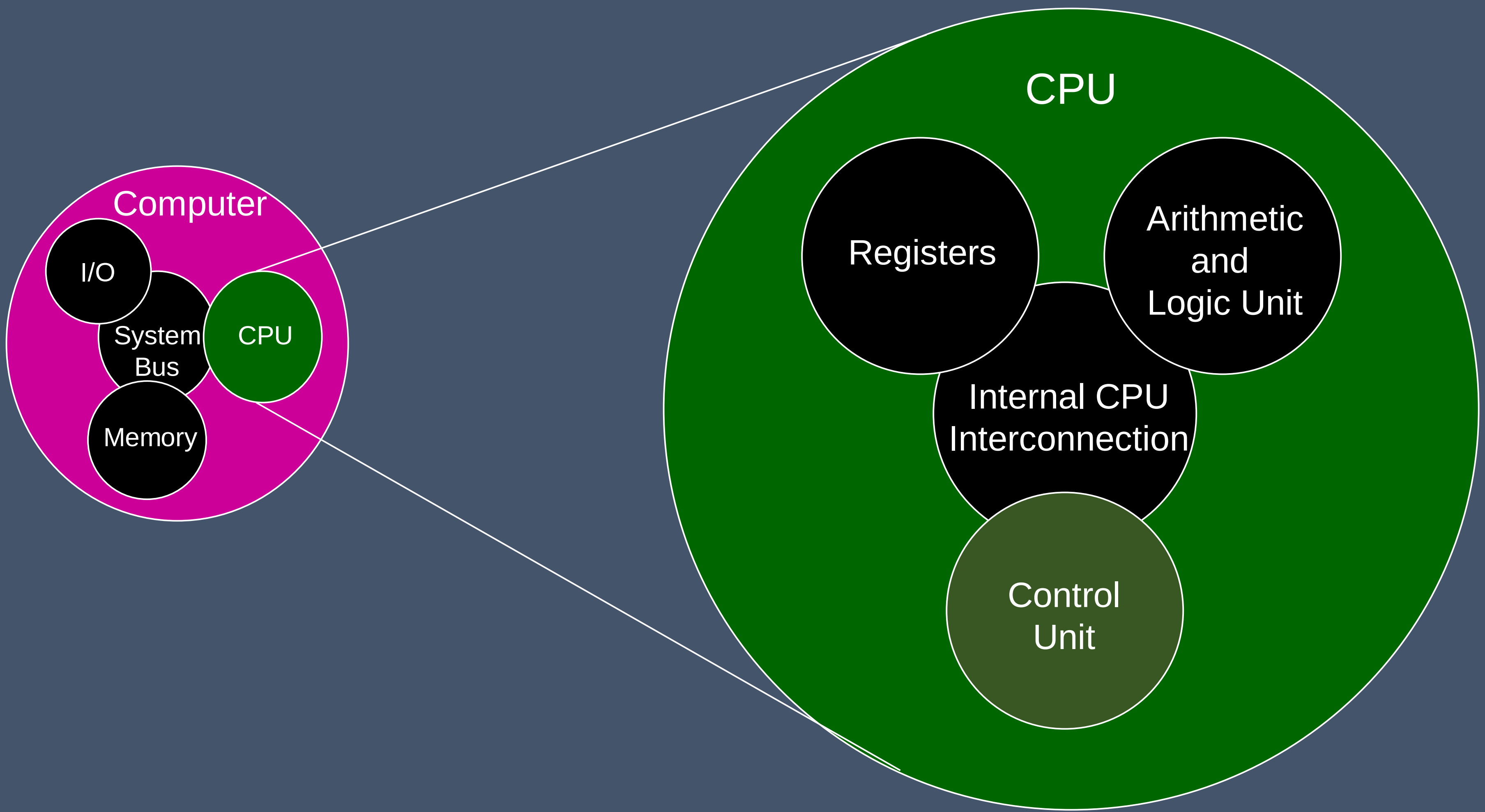


Structure
Top Level (Hardware)

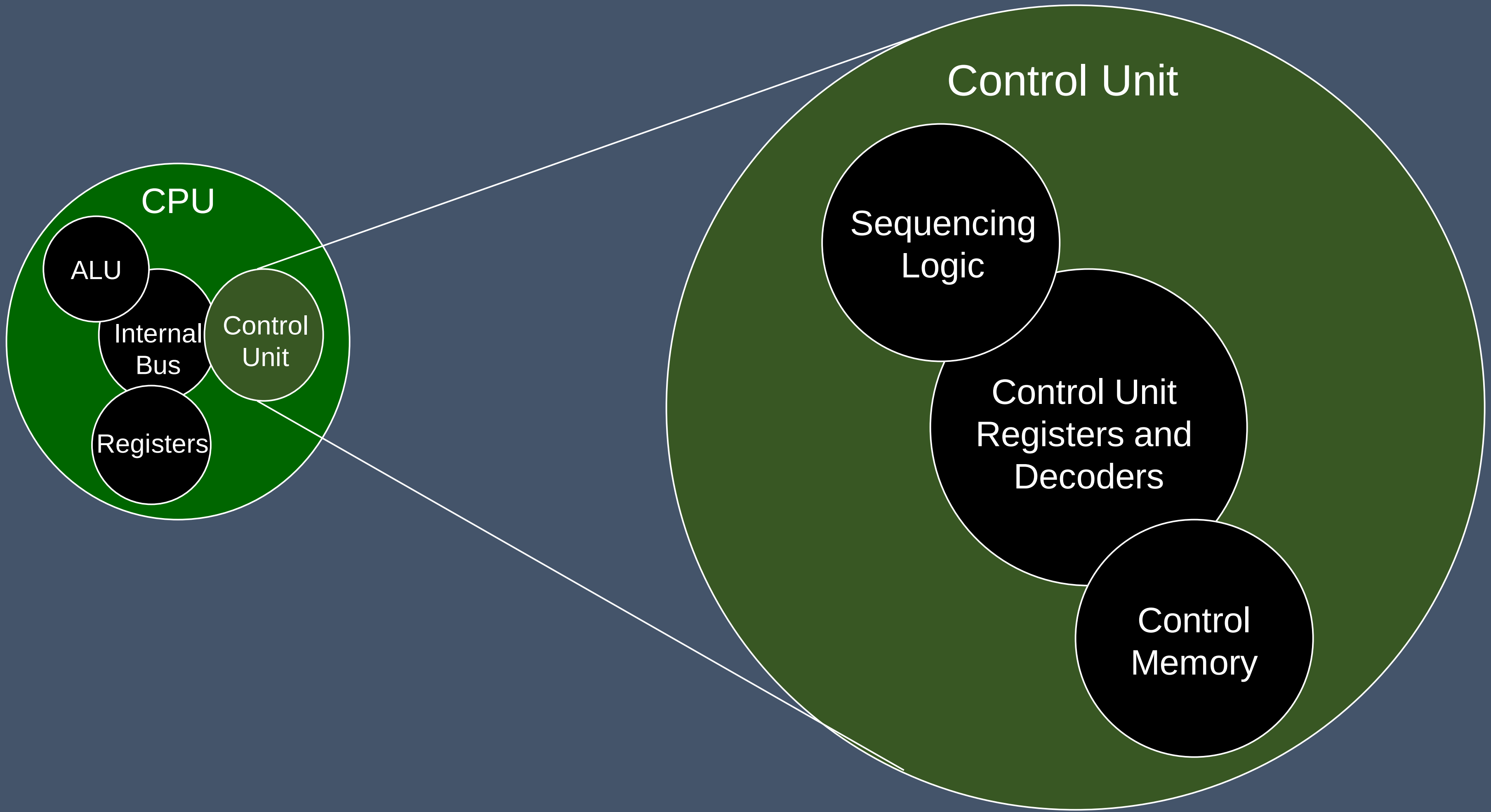
Structure
Top Level



Structure
The CPU

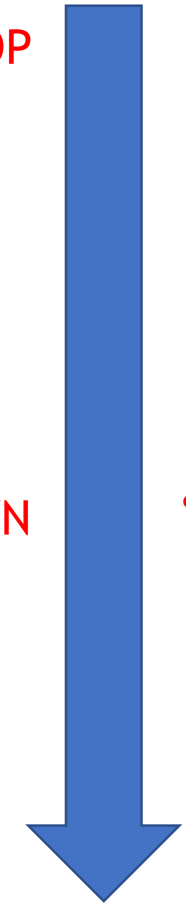


Structure
The Control Unit



Structure

TOP



DOWN

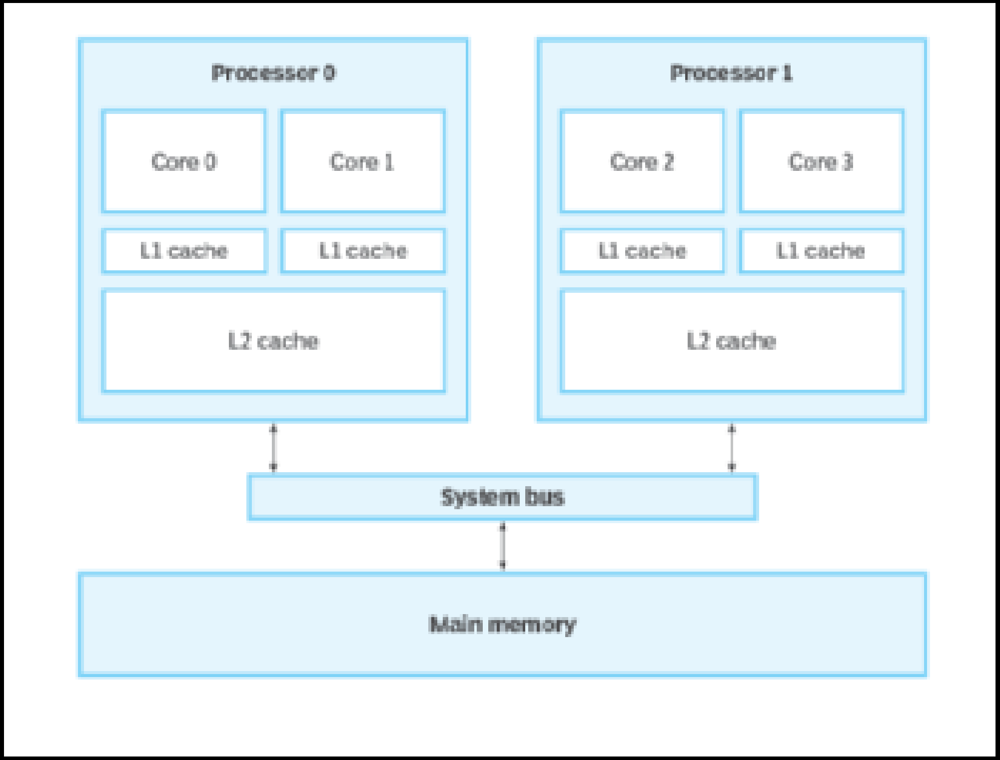
- Computer
  - Central processing unit (or CPU or processor)
  - Main memory
  - I/O
  - System interconnection

- Central processing unit
  - Arithmetic and logic unit (or ALU)
  - Registers
  - Control unit
  - Internal CPU interconnection

What can you notice?

Multicore Computer Structure

- Contemporary computers generally have multiple processors
- When these processors all reside on a single chip, the term multicore computer is used, and each processing unit (consisting of a control unit, ALU, registers, and perhaps cache) is called a core



Multicore Computer Structure

Terminology

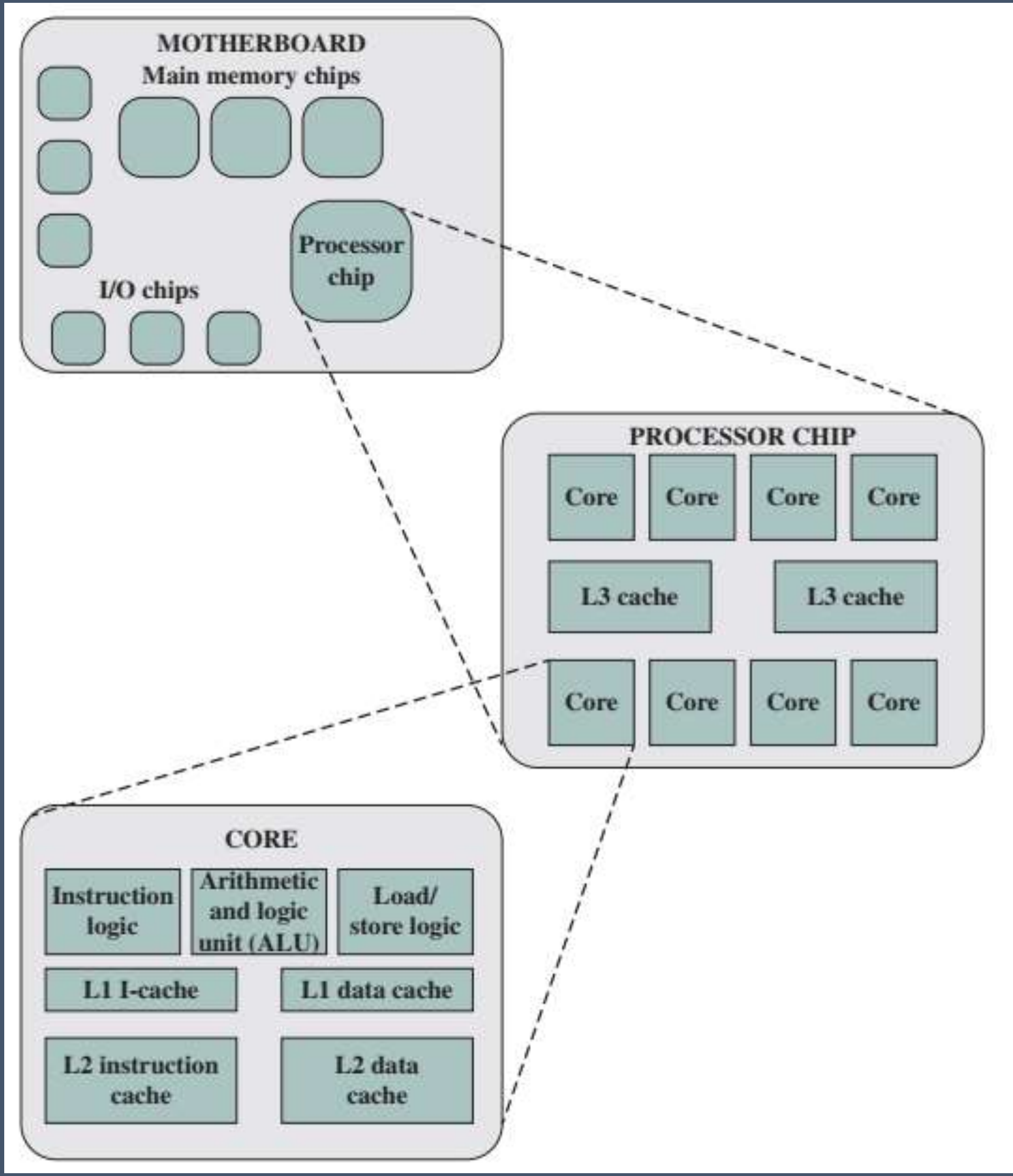
- Central processing unit (CPU)
  - Fetches and executes instructions
  - Consists of an ALU, a control unit, and registers
  - Referred to as a processor in a system with a single processing unit
- Core
  - An **individual processing unit** on a processor chip
  - May be equivalent in functionality to a CPU on a single-CPU system
  - Specialized processing units are also referred to as cores

Multicore Computer Structure

Terminology

- Processor
  - Containing **one or more cores**
  - Interprets and executes instructions
  - Referred to as a multicore processor if it contains multiple cores

Multicore Computer Structure



Reflection…

From the lecture today,
I learnt…

Link on eLearn / Chat

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